# AI-Based Energy Theft / Abnormal Consumption Pattern Detection

## Second-Year EEE Capstone Project

Beginner-friendly project report with Python, Isolation Forest, anomaly scoring, evaluation, and Streamlit dashboard.

*Educational prototype — anomaly flags are screening indicators, not proof of electricity theft.*

# 1. Abstract

This project develops an educational machine-learning pipeline that screens electricity meter readings for unusual consumption intervals. It creates simulated hourly readings, cleans the data, engineers time and history features, trains an Isolation Forest model, calculates anomaly scores, evaluates flags against simulated labels, and presents results in a Streamlit dashboard.

# 2. Problem Statement

Manual review of large numbers of electricity readings is difficult. Unusual patterns can arise from genuine changes in household activity, weather, meter faults, data errors, or possible unauthorized activity. The system prioritizes readings for review; it does not determine theft.

# 3. Objectives

Create a reproducible dataset; preprocess and explore data; engineer time-based features; train an anomaly model; display scores and flagged intervals; calculate precision, recall, F1-score and false-positive rate; build a dashboard; document responsible-use limits.

# 4. Tools and Technologies

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Streamlit and Joblib. Run the scripts locally or in a Python environment that supports these packages.

# 5. Dataset and Assumptions

The dataset contains simulated hourly meter readings, meter IDs, temperature, timestamps and injected abnormal spikes or drops. The simulated label is used only for evaluation and is excluded from training features. Synthetic labels are not real theft records.

# 6. Preprocessing and Feature Engineering

The script parses timestamps, converts consumption to numeric values, removes rows missing essential fields and negative consumption values, sorts records by meter and time, and fills missing feature values with training-set medians. Features include consumption, temperature, hour, weekday, month, weekend indicator, rolling 24-hour mean, difference from the rolling mean, and ratio to meter median.

# 7. Machine Learning Method

Isolation Forest is an unsupervised anomaly detector. Unusual points tend to be isolated by fewer random splits. The model is fitted on earlier timestamps and evaluated on later timestamps. The contamination value is a thresholding assumption and should be tuned with validation data.

# 8. Evaluation Results

The following values were produced by the included script on its generated synthetic test set. They demonstrate the workflow, not real-world theft-detection accuracy.

| Metric | Value |
|---|---|
| Test readings | 6,912 |
| Known simulated anomalies | 88 |
| Flagged readings | 445 |
| Accuracy | 0.947 |
| Precision | 0.184 |
| Recall | 0.932 |
| F1-score | 0.308 |

| | |
|---|---|
| False-positive rate | 0.053 |
| TP / FP / TN / FN | 82 / 363 / 6461 / 6 |

# 9. False-Positive Analysis

A false positive is a normal-labelled reading that was flagged. It may reflect visitors, holidays, weather-driven cooling, a change in routine, noisy readings or an oversimplified synthetic baseline. A review process should inspect nearby readings and legitimate explanations before any action.

# 10. Dashboard

Run `streamlit run app.py` after training. The dashboard offers meter filters, a consumption trend, flagged intervals, anomaly scores, and evaluation metrics.

# 11. Limitations and Responsible Use

Synthetic data may not represent real utility data. An anomaly is not proof of theft. Use only authorized, privacy-protected readings. Validate on representative labelled data before deployment. Require qualified human review and an appeal process. Never disconnect service, accuse a customer or impose a penalty solely from this model.

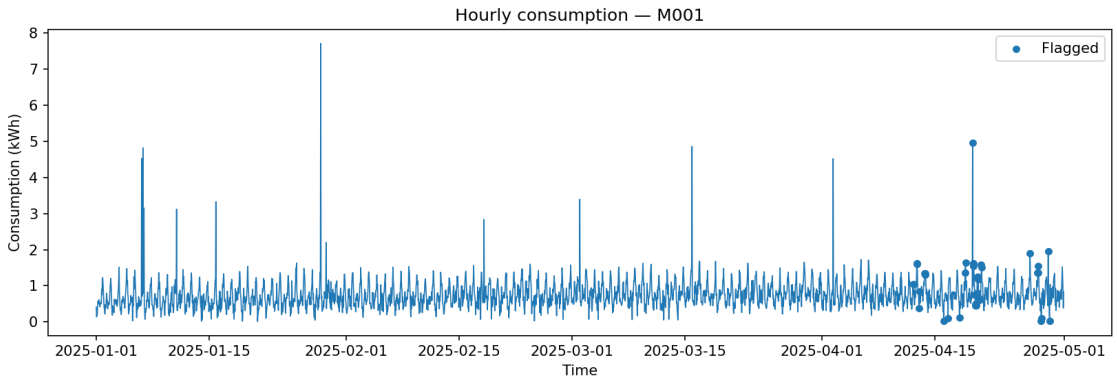
# 12. Future Scope

Test on a reputable public dataset; compare Isolation Forest with other models; add feeder-level energy balance only where authorized data exists; test across seasons and meter types; track model drift and versions.

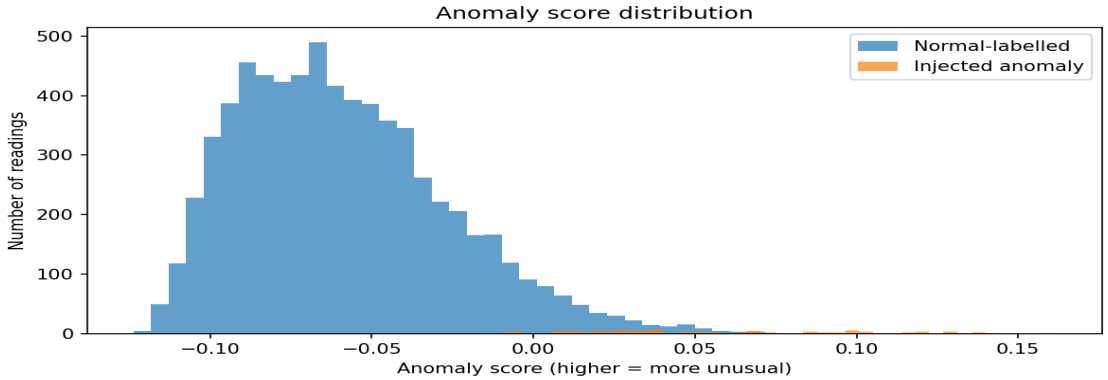
# 13. Conclusion

The project demonstrates a complete learning workflow for anomaly screening: data generation, preprocessing, feature engineering, model training, evaluation, visualization and dashboard development. It is a decision-support prototype, not a standalone theft-verification system.

# Consumption Trend




# Anomaly Scores




## Confusion Matrix

